Automation Risk, Redistribution, and Economic Vulnerability: Evidence from the European Social Survey, 2002–2022

Ignacio Urbina[1]

[1]*Stony Brook University*

September 24, 2026

Abstract

Does occupational exposure to automation risk increase demand for government redistribution? While early studies established a positive link, they left the role of institutional context theoretically ambiguous and empirically untested beyond 2012. We develop a formal insurance model of redistribution preferences that yields strict directional predictions for how welfare-state institutions moderate this risk–redistribution link. Testing these predictions on harmonized European Social Survey data spanning two decades (2002–2022)—comprising more than 270,000 respondents in 35 countries—we uncover a striking temporal reversal: the canonical positive association replicates precisely through 2012, but completely disappears in the aggregate post-2014. Decomposing this decline reveals that the automation effect was never uniform, but highly conditional on domestic institutions. Exactly as the formal model predicts, benefit earnings-relatedness amplifies the risk–redistribution gradient, flat-rate generosity buffers it, and net income inequality flattens it. Ultimately, the political consequences of technological displacement are not a uniform feature of post-industrial economies, but depend fundamentally on how domestic institutions distribute the resulting economic strain.

Main-text word count: 8,070

1 Introduction

Advances in artificial intelligence have renewed debate over the political consequences of labor-market displacement. Comparative political economists study how technological change influences policy preferences and political behavior by linking measures of occupational automation risk to public opinion data. In one important study using European Social Survey (ESS) data from 2002 to 2012, Thewissen and Rueda (2019) found that respondents in occupations with greater routine-task automation risk expressed more support for redistribution. This finding has prompted follow-on work (Im et al., 2019; Kaihovaara and Im, 2020; Gallego et al., 2022b; Heinrich and Witko, 2022). Yet that work has not examined whether automation displacement risk still translates into higher demand for redistribution, nor what institutional conditions mute or amplify that link. This paper takes up both questions. If welfare states differ in how much of that risk they insure, then the exposure–redistribution link should differ with them. That variation is the paper's subject.

Existing accounts agree that economic insecurity can raise demand for protection (Margalit, 2013, 2019), but disagree or stay silent on which institutional levers strengthen or weaken the risk–redistribution link (Moene and Wallerstein, 2001; Iversen and Soskice, 2001; Korpi and Palme, 1998). Extending the standard models of redistribution preferences, we treat redistribution demand as demand for insurance. Institutions determine how much income a worker keeps after displacement due to job automation. A worker facing displacement risk values the tax-and-transfer system as a hedge, so how strongly automation exposure moves that demand depends on how much of the loss existing institutions already cover, and in what form. Starting from a simple formal model of redistributive taxation as individual demand for insurance, this paper derives predictions about the institutional factors that moderate the association between automation exposure and demand for redistribution. The model's comparative statics for institutions depend on one condition: whether the income a worker keeps after displacement is proportional to prior earnings. Flat transfers weaken the effect of automation risk on redistribution demand, while benefits that track prior earnings strengthen it; employment protection enters through displacement risk, and net inequality through relative income.

Institutions differ across countries and change over decades, so a test needs both kinds of variation. We therefore rebuild the benchmark design from raw ESS records harmonized to a common format for eleven waves between 2002 and 2022, spanning 35 countries and extending coverage a decade beyond the published window. Both measures we use operationalize the same core idea: routine, rule-following work is the easiest for digitization and machine automation to displace. We use the occupation-level automation probability of Frey and Osborne (2017) as our reference index. For robustness, we also test the routine-task-intensity index of Goos et al. (2014), which serves as the closest analog to the canonical Thewissen–Rueda benchmark. We estimate country and country-year slopes from a single multilevel model and examine whether perceived vulnerability links occupational exposure to redistribution preferences and whether welfare-state and labor-market institutions condition that link.

The benchmark estimate of Thewissen and Rueda (2019) replicates in our harmonized data over the original period. Refitting the published estimator on the original replication data recovers the baseline almost exactly, and the positive association reappears in the harmonized panel over 2002–2012. Yet, under every country roster in the analysis, the post-2014 aggregate exposure–redistribution association is indistinguishable from zero. This decline is estimated directly rather than read off subsamples. Strikingly, the post-2014 attenuation is larger in absolute magnitude than the pooled historical gradient itself. Neither changing country coverage nor sample composition explains this drop. While measured routine exposure itself fell in the benchmark countries over this period—and some of the attenuation may stem from occupational measurement error after the ISCO-08 revision (van Hoorn, 2022)—the data reveal that variation across countries far exceeds variation over time.

Automation exposure predicts financial hardship in the same waves in which it predicts redistribution support: on the full sample the strain gradient is significant in six of the seven waves where the redistribution gradient is, and in none where it is not. Exposure also predicts expected job loss and perceived job insecurity in the rounds that include them. Net of the business cycle, institutions moderate the exposure–strain link between countries rather than within them, and the gradient runs flatter where flat-rate transfer generosity is high and steeper where employment protection and net inequality are high. When the four focal moderators are estimated

jointly, all three signed predictions on redistribution are supported. net inequality flattens the risk–redistribution gradient within countries, benefit earnings-relatedness amplifies it between countries, and flat generosity buffers it within countries, while the employment-protection prediction receives no support. The opposite signs of the two transfer routes match the model's prediction: earnings-related benefits strengthen the exposure–redistribution gradient, and flat transfers weaken it.

Related work has documented automatability–redistribution associations in the ESS with other exposure measures and designs (van Hoorn, 2022), policy-specific responses to technological risk (Gallego et al., 2022a; Busemeyer and Tober, 2023), welfare-state attitudes under automation risk (Busemeyer and Sahm, 2022), and the electoral response to state automation policy (Gingrich, 2019). Busemeyer and Tober (2023) report institutional moderation of the opposite sign to the one predicted here, and Jeffrey (2020) find redistribution preferences flat or falling under randomised automation vulnerability. Against that literature, this paper makes three contributions. The first is a theoretical correction. The benchmark literature relies on an informal assumption that income and risk interact positively to drive redistribution demand. Our formal model demonstrates that this interaction contradicts standard estimates of risk aversion; it is a restriction on parameters rather than a natural implication of insurance logic, and our model formally rules it out.

The second contribution clarifies how the specific architecture of a welfare state shapes political coalitions. Since Korpi and Palme (1998), political economists have often treated flat-rate transfers and earnings-related benefits as varying points on a single spectrum of welfare generosity. Our model proves they have opposing political consequences. Flat transfers provide a universal floor that buffers economic anxiety, which flattens the political demand for redistribution. In contrast, benefits that scale with a worker's prior earnings preserve income hierarchies, meaning that high-earning professionals facing automation still have vastly more to lose. By demonstrating that these two designs operate in strict mathematical opposition, we show that a country's exact benefit structure fundamentally dictates its risk politics.

The third is that the benchmark association was never general. Earlier work estimates one pooled coefficient and the moderation literature explains that composite. Decomposing it into

country-year slopes shows the association strong in some countries, fading in most during the 2010s, and never appearing at all in others. That finding rests on the exposure indices meaning the same thing across countries and waves, an assumption the attenuation results above put under strain.

The paper proceeds as follows. The theory section develops the insurance model and derives the hypotheses, with the derivations in Appendix 8.1. The research design section describes the panel, the exposure measures, and the estimator. The results section reports the benchmark replication, the post-2014 attenuation, and the country-year slopes. The mechanism section traces exposure to perceived vulnerability and tests the four institutional predictions. A discussion and a conclusion follow. Routine-task-intensity results and further robustness checks are in the appendix.

2 Theory

2.1 Comparative Political Economy of Redistribution

Our starting point is two accounts of welfare-state development and redistribution. Wilensky's modernization account treats welfare-state expansion as a response to industrialization, economic growth, and demographic change (Wilensky, 1974). Stephens' power-resources account instead locates the sustaining force in organized labor and its political arm, from union strength to socialist parties and collective bargaining (Stephens, 1979). These accounts distinguish the creation of economic risk from its political consequences: institutions and class power shape how people experience new risks.

The comparative political economy literature also clarifies why redistribution demand need not mechanically follow from objective exposure. Moene and Wallerstein (2001) show that preferences over social insurance and redistribution depend on the distribution of risk and income, not on inequality alone. Iversen and Soskice's asset theory similarly treats social-policy preferences as a function of exposure to labor-market risk and the portability of workers' skills (Iversen and Soskice, 2001). Korpi and Palme's paradox of redistribution adds an institutional feedback: encompassing welfare states can reduce poverty and inequality because broad social programs

build larger political coalitions for redistribution (Korpi and Palme, 1998). These arguments imply that automation exposure should have political consequences when it is experienced as uninsured labor-market risk, and that welfare-state and labor-market institutions should condition that experience.

2.2 Job Automation and Economic Threat

Automation can alter perceptions of economic threat, which in turn can change political attitudes. A direct literature now links occupational automatability to policy preferences: van Hoorn (2022) documents a positive automatability–redistribution association in the ESS with an occupation-level risk measure, Gallego et al. (2022a) find that technological risk shapes policy preferences in policy-specific ways, Busemeyer and Tober (2023) show the response depends on institutional context, and Heinrich and Witko (2022) connect technology threat to candidate preferences in the United States (see also Im et al., 2019; Kaihovaara and Im, 2020; Gallego et al., 2022b). Research on economic shocks links perceived insecurity, especially anticipated unemployment, to changes in political and social preferences (Margalit, 2019, 2013). A related literature on economic vulnerability and political psychology likewise treats perceived threats as a potential source of support for protective policies (Hacker et al., 2013; Rogers, 1975). Automation is one potential source of such disruption: people in exposed occupations may regard their employment prospects as less secure (Autor and Salomons, 2018).

2.3 Automation Risk as Conditional Redistribution Demand

The benchmark claim is that routine-task automation risk increases demand for redistribution because occupational exposure can signal future income and employment losses (Thewissen and Rueda, 2019). Respondents in routine, automatable occupations may anticipate displacement or downward mobility, making redistribution and social insurance more attractive. We therefore expect automation exposure to be positively associated, on average, with support for redistribution.

The association need not be constant over time. As economies develop, employment structures and institutions change, and with them how occupational risk is understood. Reces-

sions, recoveries, and inflation can make the same exposure feel more or less threatening. Because people and firms also adapt as automation diffuses, through changed expectations and tasks that shift across occupations, routine-task measures may lose political relevance as technological change advances beyond routine-biased computerization. We expect the automation–redistribution association to vary across survey waves rather than to move monotonically in one direction.

The same exposure may also have different consequences across countries. Welfare states and labor-market institutions shape whether automation risk is privately borne or collectively insured. Where workers are strongly organized and bargaining is broad, they hold power before market risk becomes individualized, and generous unemployment protection, social spending, and redistribution can buffer the material and psychological consequences of exposure. Where such protection is weaker or inequality is higher, exposure may generate greater demand for redistribution because the risk is less insured.

We use perceived vulnerability to trace how exposure may translate into redistribution demand. Exposure may appear first in expectations of job loss, financial strain, and insufficient income. We therefore advance three hypotheses: H_1 expects higher occupational automation risk to predict more support for redistribution, H_2 expects that association to vary across time, and H_3 expects it to vary across countries. Beyond the hypotheses, M_1 asks whether perceived economic vulnerability partly links automation risk to redistribution demand, and M_2 asks whether welfare-state and labor-market institutions moderate the association between exposure, vulnerability, and redistribution.

2.4 An Insurance Model of Redistribution Demand

A worker who expects to keep her job values a tax-and-transfer program for what it pays her now, and a worker who expects to lose it values the same program for what it would pay her then. Automation exposure acts on the second valuation, and the model below makes that channel explicit enough to sign, with the algebra in Appendix 8.1.

Write her income relative to the mean as ξ, and let p be her displacement probability. Displacement is a temporary spell, and her income during that spell depends on four quantities

that welfare states set separately. These are the expected share of the period out of work, μ, the re-entry wage as a fraction of her old wage, ρ, the earnings-related benefit rate, b, and the benefit duration, B. Her retention share δ collects them into one number,

$$\delta \;=\; (1-\mu)\,\rho \;+\; b \cdot \min(\mu, B). \tag{1}$$

The first term is her post-spell wage income, scarred by the interruption, and the second term is her benefit income while the entitlement runs. The retention share is therefore not a free parameter for protection in general, but a quantity with four separate components, each of them the product of a different institution. An index cannot say which component produced the result.

A linear tax τ pays for transfers, of which she receives θ_g in work and θ_b out of work. Payments proportional to her own past earnings belong to δ, and they reach her whatever tax rate she votes for. Payments proportional to the tax rate belong to θ_b, and they pay her the same amount as everyone else. Earnings-related unemployment insurance therefore falls on one side of this distinction and social assistance on the other, however alike their generosity looks in a spending series.

Her preferred tax rate weighs the two states by their probabilities and by the sharpness of her distaste for the gap between them, which the model writes as risk aversion σ. Maximising expected utility gives a closed form,

$$\tau^{*} \;=\; \frac{\xi\,(1-\kappa\delta)}{(\xi-\theta_g)+\kappa\,(\theta_b-\delta\xi)}, \qquad \kappa \;\equiv\; \left[\frac{(1-p)\,(\xi-\theta_g)}{p\,(\theta_b-\delta\xi)}\right]^{1/\sigma}, \tag{2}$$

where κ is the ratio of her consumption in one state to her consumption in the other. That preferred rate is her demand for redistribution, and it is the disposition the ESS item measures when it asks whether government should reduce differences in income levels. A rise in τ^{*} is a rise in support for redistribution, and the exposure–redistribution gradient below is the change in τ^{*} per standard deviation of automation exposure.

The retention share reaches her demand through that ratio. A higher δ leaves her more income after displacement, so the two states move closer together. But it does not simply calm her, because a worker with earnings-related cover still has something proportional to lose as

her risk rises. A flat transfer does the opposite and lifts the floor under everyone by the same amount, so its value is largest for the worker with least at stake and smallest for the worker with most. Two countries can therefore sit on opposite sides of this prediction on identical social spending, and a single replacement rate cannot tell them apart.

One proposition states the model's unconditional prediction, and three more sign the four institutional moderators that Table 1 takes to the data. The appendix gives the proofs.

> *Risk raises demand unconditionally.* A worker's preferred tax rate rises with her displacement probability, for every level of risk aversion and every setting of the remaining parameters (Proposition 2).

This is the model's only unconditional prediction, and it needs no assumption about how cautious she is, because a higher chance of displacement puts more of her expected life in the state where transfers pay her and less in the state where she funds them, so the rate she would choose rises. H_1 applies this prediction to occupational exposure, on the assumption that exposure raises her perceived chance of displacement with the wage and severity channels fixed. No institution enters here, and Table 1 therefore does not list it.

The four institutional predictions follow, in the order the table reports them, and each one states the moderator it signs and the proposition behind it.

> *Benefit earnings-relatedness amplifies the gradient.* Exposure raises her preferred tax rate more where a larger share of a displaced worker's former earnings is replaced (Proposition 8).

> *Flat generosity buffers the gradient.* Exposure raises her preferred tax rate less where transfers pay the same sum regardless of former earnings (Proposition 8).

Income proportionality divides these two. A flat transfer pays the same sum whatever she used to earn, so it matters most to the worker with least at stake and it compresses the gap that exposure works on. Earnings-related cover keeps her loss proportional to her old wage, so a worker with more to lose still has more to lose, and exposure keeps its grip. The same proposition signs both, because an identity in the model ties the two derivatives together so that a country cannot come out on the wrong side of both, which makes the pair a single test rather than two separate ones.

> *Employment protection steepens the gradient, but only above log utility.* Exposure raises her preferred tax rate more where the baseline chance of displacement is lower, provided risk aversion exceeds one. At log utility the effect is exactly zero (Propositions 3 and 4).

This result concerns curvature, meaning how the exposure slope itself changes as baseline risk changes, and the panel does not estimate that curvature directly. A linear cross-level interaction asks whether the slope differs between countries with stricter and looser protection, so employment protection enters as an institution that shifts baseline risk and the interaction tests the sign the curvature result implies rather than the curvature itself. The reasoning is that a worker who already expects to lose her job has little left to learn from a further rise in risk, while a worker who expects to keep it revises sharply, and every country in this panel sits below the displacement probability of one half at which that logic holds. An institution that lowers the baseline therefore steepens the gradient without changing the retention share at all. The channel is second order, however, and it disappears at the risk aversion near one that the income results invoke, so the model predicts no employment-protection effect at that calibration.

> *Net inequality flattens the gradient.* Exposure raises her preferred tax rate less where relative income is more dispersed (Propositions 5 and 6).

The model does not sign the income effect in general, because two forces oppose each other and risk aversion decides the balance. At high risk aversion insurance behaves as a normal good and richer workers buy more of it, the region Moene and Wallerstein (2001) describe. At low risk aversion the redistributive motive dominates and richer workers want less, the benchmark of Meltzer and Richard (1981). Meta-analytic estimates put relative risk aversion near one, which places this panel in the second region, so rising inequality crowds the insurance motive out and the prediction is a flatter gradient.

Risk in this model is a perception, so exposure ought to appear in her account of her own position before it appears in her demands on government. M_1 tests that link and M_2 tests the four institutional results above.

2.5 Measuring Routine-Task Automation Risk

The measures used in this literature rate occupations by the routineness of their task content, because computers and machines can more readily perform codifiable, repetitive tasks (Autor and Salomons, 2018). We use the engineering-bottleneck probability of Frey and Osborne (2017) and the routine-task-intensity (RTI) index of Goos et al. (2014), which correlate at 0.595 in our

sample. Both measures are highest in clerical, production, and lower-skill service occupations, where pay and employment security also tend to be lower.

This is the construct used in the redistribution literature, and we retain both operationalizations throughout the paper. If perceived vulnerability tracks labor-market precarity, routine-task exposure should predict greater vulnerability among respondents in the occupations it identifies as exposed. The automation–redistribution association should be strongest where institutions protect least against that vulnerability.

3 Research Design

3.1 Data and Measures

The analysis uses a harmonized panel of eleven European Social Survey (ESS) waves from 2002 to 2022, spanning 35 countries and roughly 270,000 respondents before listwise deletion. The outcome is the standard ESS item asking whether government should reduce income differences. It is coded on a five-point scale, with higher values indicating greater support for redistribution. Following Thewissen and Rueda (2019), the main analysis is limited to employed and unemployed members of the active labor force.

Figure 1: Distribution of Job Automation Risks in the ESS Sample

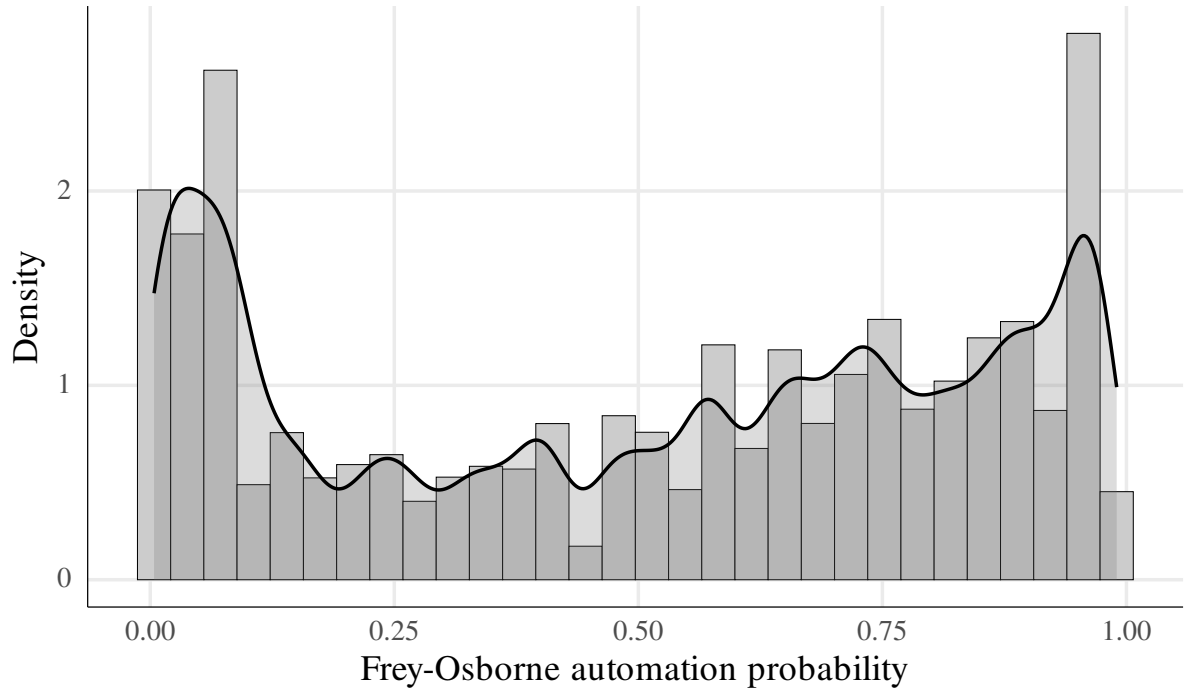


Figure 1 shows Frey–Osborne automation probabilities for ESS occupations.[1] Because this measure is a probability, observations cluster toward the ends of the 0–1 scale. Its density resembles the corresponding U.S. distribution in Frey and Osborne (2017), which is consistent with a successful occupation match.

Frey–Osborne is our reference index, and alongside it we use the Goos–Manning–Salomons routine-task-intensity (RTI) index as a second, independently constructed measure of the same routine-task construct. The panel spans the change in ESS occupation coding: rounds 1–5 use ISCO-88 and rounds 6–11 use ISCO-08, so we harmonize both exposure measures across the two occupation systems.

We measure economic position with income that is equivalized for household size and expressed relative to the country-year mean. We convert income-band responses in ESS show cards to monetary midpoints, and for the open top band we estimate a Pareto mean from the observed counts in the top two categories, following Hout (2004). We then apply the square-root equivalence scale and divide the resulting value by the design-weighted country-year mean. This control matches the baseline specification in the benchmark literature.

[1] The match covers almost 90% of potential cases; matched and unmatched cases do not differ systematically.

We validated the harmonization by joining the raw panel to the original replication data of Thewissen and Rueda (2019) at the respondent level for rounds 1–6. RTI matches exactly, Frey–Osborne probabilities match almost exactly, and the redistribution outcome and core demographic variables have near-perfect agreement. These checks support treating the 2002–2022 panel as a direct extension of the published benchmark design.

The mechanism analyses use several indicators of economic vulnerability. Financial strain is available in every wave and is coded as whether the household finds it difficult or very difficult to cope on its current income. Selected waves provide additional outcomes: expected job loss (a binary indicator for expecting to lose one's job in the next twelve months) and expected income insufficiency (a binary indicator for expecting income too low to meet needs) in the 2008 and 2016 welfare-attitudes modules, and perceived job insecurity (a scaled measure of the perceived likelihood of job loss) in the 2004 and 2010 work modules, all restricted to labor-market participants.

At the country level, we examine 11 institutional moderators. The Comparative Political Data Set supplies six annual series (CPDS; Armingeon et al., 2025): union density and collective-bargaining coverage (interpolated from the OECD/AIAS ICTWSS series), employment-protection strictness, left cabinet share, industry employment share, and trade openness. Four additional series come from their source APIs: public social spending as a share of GDP (OECD SOCX), the replacement rate for net unemployment benefits (OECD tax–benefit models; a single person without children at 67% of the average wage, six months into unemployment, without social-assistance top-ups), net income inequality, and relative redistribution (SWIID, version 9.92). The eleventh moderator, benefit earnings-relatedness, measures how far unemployment benefits track prior earnings: it is the difference in the OECD net replacement rate between a claimant at the average wage and one at 67% of it, at month six of a spell, a value near zero means benefits are fully earnings-related, and a large negative value means they are capped or flat-rate. The generosity series cannot separate flat-rate from earnings-related systems because it is evaluated at a single earnings level, and the insurance model predicts the two designs moderate the gradient in opposite directions. Fetscher (2023) measures that structure from replacement rates across earnings levels, deriving a benefit-concentration indicator from the same OECD calcula-

tor over rank-ordered income groups, in the tradition of Beramendi and Rehm (2016) and the progressivity index of Kakwani (1977). Ours is a deliberately coarse two-point version, chosen so the series runs annually for every country-year in the panel. It is also narrower. A window closing at the average wage understates earnings-relatedness wherever a statutory ceiling is the binding feature, and ceilings bind above the average wage in many systems. The two-stage diagnostic uses their 2016 values, contemporaneous with the country slopes they are correlated with. The moderation models use country-year values that are standardized and lagged by one year. Depending on availability, those models contain 25–35 national clusters.

3.2 Empirical Strategy

For redistribution preferences, we estimate one multilevel model with random effects for each exposure index. It regresses individual support for redistribution on occupational automation exposure and the benchmark controls. The automation-risk slope has global, country, year, and country-year components (equations 3–4). Every reported redistribution result is derived from that fit, without re-estimating any coefficient on subsamples. The country and year profiles are random-effect components, the country-year map is the composite $\beta_{c,t}$, and the replication, composition, and institutional diagnostics aggregate or correlate those fitted slopes.

To ensure consistency with the published benchmark, our primary models include the baseline control vector established by Thewissen and Rueda (2019). This vector controls for relative equivalized income, years of education, gender, age, current or former union membership, religiosity, and individual unemployment status, adjusted using ESS design weights and accounting for the country-year structure of the survey. The appendix probes the gradient against rival individual-level explanations one at a time, in the style of the benchmark study: left–right self-placement, occupational unemployment risk, contract precarity, household size, and one-digit occupation dummies, with a simultaneous specification admitted under a stated correlation and variance-inflation rule (Table 16), accompanied by an individual-level correlation table (Table 17).

We estimate the mechanism analyses separately from the redistribution models. At the individual level, we decompose the association between occupational exposure and each perceived-

vulnerability outcome into paths through relative income, occupational unemployment risk, and contract precarity. These are statistical path decompositions in a single cross-section: exposure is not assigned, the channels are themselves products of occupational sorting, and we make no sequential-ignorability assumption, so the decomposed paths describe how the association is distributed across measured channels rather than identify causal mediation. At the country level, each institutional measure is interacted separately with individual automation risk. Models using multiple waves include one-year-lagged moderators, year fixed effects, exposure slopes that vary by country, and a country-year random intercept. The intercept absorbs wave-level macro shocks that would otherwise understate the standard errors of interactions across levels in pooled cross-sections (Schmidt-Catran and Fairbrother, 2016). These models assess whether protective institutions weaken the association between occupational risk and perceived vulnerability.

Let y_{ict} denote the outcome for individual i in country c in survey year t, and let AR_{ict} denote the occupational automation-risk index. Depending on the analysis, y_{ict} is support for redistribution (the primary outcome), financial strain, expected job loss, expected income insufficiency, or perceived job insecurity. The analytic specification we use for all models is:

$$y_{ict} = \alpha_{ct} + \beta_{ct} AR_{ict} + \mathbf{X}_{ict}\boldsymbol{\phi} + \varepsilon_{ict}, \tag{3}$$

where α_{ct} is a country-year-specific intercept, $\mathbf{X}_{ict}$ is the vector of individual-level controls with coefficient vector $\boldsymbol{\phi}$, and β_{ct} is the country-year-specific slope for automation risk. These slopes are modeled as:

$$\beta_{ct} = \gamma_{10} + u_{1c} + v_{1t} + \omega_{ct}, \tag{4}$$

where γ_{10} is the global baseline slope, u_{1c} and v_{1t} are country and year deviations, and ω_{ct} is the country-year interaction. ε_{ict} is the individual-level error; dependence within country-years is absorbed by the country-year components of the random part rather than by a post-estimation clustering correction, and all reported standard errors are model-based maximum-likelihood throughout. Where figures display the fitted slope components themselves, their 95% intervals combine the sampling variance of the fixed slope with the conditional variances of the relevant

country, year, and country-year components, ignoring cross-component posterior covariances, the standard approximation for intervals around best linear unbiased predictions. We estimate this model once per exposure index on the full labor-force sample from 2002 through 2022, using design weights and standardizing exposure on the pooled estimation sample.

The other analyses are instances of the same specification. The vulnerability models replace the outcome with each perceived vulnerability indicator and estimate equation 3 as design-weighted linear probability models with random intercepts; the mediation-style decompositions use a cluster bootstrap that resamples countries with replacement, with 2,000 replications. The institutional moderation models take the within–between form developed for repeated cross-sections of respondents nested in countries observed over multiple waves (Bell and Jones, 2015; Fairbrother, 2014; Bell et al., 2019) and extended to cross-level interactions by Giesselmann and Schmidt-Catran (2019):

$$y_{ict} = \alpha_{ct} + \beta_c\, AR_{ict} + \gamma_B\big(AR_{ict} \times \bar{Z}_c\big) + \gamma_W\big(AR_{ict} \times (Z_{c,t-1} - \bar{Z}_c)\big) + \mathbf{X}_{ict}\boldsymbol{\phi} + \lambda_t + \varepsilon_{ict}, \quad (5)$$

where $Z_{c,t-1}$ is the lagged institutional moderator, $\bar{Z}_c$ is its country mean, and the within-country variation comes from the deviation $(Z_{c,t-1} - \bar{Z}_c)$. The coefficient γ_B measures between-country moderation and γ_W measures within-country moderation. Main effects of both moderator components, exposure interactions with contemporaneous unemployment and real GDP growth, year fixed effects λ_t, exposure slopes that vary by country, and a country-year random intercept complete the specification.

The moderators divide by their status in the insurance model. Four are focal because a stated proposition signs them, each through one primitive of the model. Proposition 8 signs benefit earnings-relatedness through the retention share δ and predicts it amplifies the gradient. A flat-generosity bundle, the mean of standardized social spending and realized redistribution, maps to the flat transfer θ_b, and the same proposition predicts it buffers. Employment-protection strictness maps to displacement risk p, and the curvature result of Proposition 3 predicts it steepens. Net income inequality maps to relative income ξ, where the crowding-out of the insurance motive predicts it flattens. Table 1 records the mapping. The bundle excludes the single-point replacement rate, which cannot draw the flat-versus-earnings-related distinction θ_b and δ re-

quire, a limitation of that series rather than of the underlying data (Fetscher, 2023), so it joins left cabinet share, industry employment share, and trade openness as uninteracted country-year controls in every model. Union density and bargaining coverage act on several primitives at once and the model leaves them unsigned, so the focal design excludes them; their interactions appear only in the appendix sweep. Because each proposition is a partial derivative, holding the other primitives fixed, the four focal moderators enter the primary redistribution model jointly, so each interaction is estimated conditionally on the others, with the one-at-a-time versions retained as appendix robustness. The strain models carry no signed predictions and are estimated one moderator at a time as mechanism diagnostics. Benjamini–Hochberg adjustment applies within the sixteen focal coefficients per outcome. The full symmetric sweep of all eleven moderators without institutional controls, which earlier drafts treated as primary, is reported in the appendix as robustness.

Table 1: Focal moderators and the comparative statics they test

Moderator	Primitive	Result	Predicted sign
Benefit earnings-relatedness	δ (retention)	Prop. 8	$+$ amplifies
Flat-generosity bundle	θ_b (flat transfer)	Prop. 8	$-$ buffers
Employment protection	p (displacement risk)	Props. 3–4	$+$ steepens if $\sigma > 1$; nil at $\sigma = 1$
Net inequality	ξ (relative income)	Props. 5–6	$-$ flattens
Controls: left cabinet share, industry employment, trade openness, replacement rate			

Because respondents are sampled afresh in each wave, the decomposition applies to the country panel rather than to individuals (Fairbrother, 2014). The random exposure slope guards against anti-conservative inference on cross-level interactions (Heisig and Schaeffer, 2019). The term γ_B is also the fixed-part control for between-country heterogeneity in the exposure slope that Giesselmann and Schmidt-Catran (2019) show is required for γ_W to be a genuine within estimator; a random slope alone does not serve that purpose.

Two conventions govern every significance statement in the paper. First, when the text calls a coefficient significant, the claim refers to its Benjamini–Hochberg adjusted q value at

the $q < 0.10$ threshold, computed within the coefficient's design-defined family;[2] the family sizes are stated where each family is introduced, and unadjusted standard errors are reported throughout, so a reader can apply a different rule. The registered confirmatory vulnerability table instead keeps the stricter Holm adjustment within each outcome–wave pair, preserving its registration (Table 6). Second, the figures match the testing rule they display: forest plots of adjusted families fill a symbol when $q < 0.10$ and draw unadjusted 90% whiskers, the interval whose level corresponds to that criterion, so a filled symbol never brackets zero; every other interval in the paper is 95%.

4 Attenuation and Cross-Country Heterogeneity in the Automation–Redistribution Association

We report the benchmark in three steps: an exact reproduction of the published estimator, window aggregates from our panel model, and a direct test of the change over time. Fitted on the replication data of Thewissen and Rueda (2019) with their sample restrictions and a country random intercept, their baseline returns a Frey–Osborne coefficient of 0.162 against the published 0.162 ($N = 59,997$) and an RTI coefficient of 0.042 ($N = 64,639$), so the published estimates are recovered on their own data before any harmonization of ours enters. The harmonized ESS panel then reproduces the Frey–Osborne association in the original 2002–2012 window under the benchmark outcome, labor-force restriction, relative-income control, and exposure indices. When the panel is extended through 2022, the aggregate association is smaller. The RTI results are reported in the appendix.

Table 2 summarizes the fitted Frey–Osborne slope by time window. To compare it with the published coefficient, we rescale each estimate to the probability's 0–1 range. In 2002–2012, the aggregate is 0.177 (95% CI $[0.099, 0.254]$), which brackets the published benchmark of 0.162. In

[2] The Benjamini–Hochberg procedure controls the false discovery rate: the expected share of false positives among the coefficients a family declares significant (Benjamini and Hochberg, 1995). Within a family of m tests, the raw p values are ranked $p_{(1)} \leq \cdots \leq p_{(m)}$, the largest k with $p_{(k)} \leq (k/m)\, q$ is found, and the k lowest-ranked hypotheses are rejected at level q. A coefficient's reported q is its adjusted p: the smallest level at which it would be rejected, never below the raw p. Unlike familywise corrections such as Holm's (Holm, 1979), which control the probability of even one false positive and pay for that guarantee in power, false-discovery-rate control tolerates a known expected share of false discoveries, here ten percent, across a family of related tests.

2014–2022 it is 0.041 and not distinguishable from zero, and the full-panel estimate lies between those values.

The benchmark-roster rows repeat the calculation for the seventeen countries in the published sample. They show the same pattern: the 2002–2012 aggregate is 0.147 (CI $[0.068, 0.226]$), while the 2014–2022 aggregate is 0.022 and again indistinguishable from zero. These window aggregates describe the fitted slope surface. The inferential test of the change is a single model in which exposure interacts with a post-2014 indicator, so the interaction coefficient is the attenuation and carries its own standard error. That coefficient is -0.039 per standard deviation of exposure (SE 0.005) against a pre-2014 slope of 0.049, and -0.021 (SE 0.005) for RTI. A breakpoint-free version interacts exposure with every survey year, and the joint test that all year interactions are zero gives $\chi^2(10) = 127.82$ for Frey–Osborne and $\chi^2(10) = 36.71$ for RTI. Neither replication nor attenuation, therefore, depends on the broader country coverage of the panel, and the decline is a directly estimated contrast rather than a comparison of marginal intervals. The final row gives the standardized per-SD aggregate used in the figures. These estimates indicate historical attenuation alongside national variation rather than a uniform disappearance of the relationship.

Table 2: Global FO automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	N	Countries
Published benchmark	2002–2012	native	0.162	–	64,639	17
Window aggregate of $\beta_{c,t}$	2002–2012	native	0.177	[0.099, 0.254]	103,530	33
Window aggregate of $\beta_{c,t}$	2014–2022	native	0.041	[-0.037, 0.119]	91,317	33
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.113	[0.038, 0.187]	194,847	35
Benchmark-roster aggregate	2002–2012	native	0.147	[0.068, 0.226]	69,788	17
Benchmark-roster aggregate	2014–2022	native	0.022	[-0.057, 0.102]	56,378	16
Benchmark-roster aggregate	2002–2022	native	0.091	[0.015, 0.167]	126,166	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.038	[0.013, 0.063]	194,847	35

Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark's sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

Table 3 places the pooled gradient beside canonical individual-level determinants of redistribution support, estimated in one model per exposure index. One standard deviation of Frey–Osborne exposure is associated with 0.034 scale points of support, 0.9% of the design-weighted sample mean of 3.82 on the 1–5 scale, while the RTI coefficient is 0.023. In occupational terms, one standard deviation of Frey–Osborne exposure is the distance from the two-digit group nearest the panel mean, Electrical and electronic trades workers (0.53), to the group nearest one standard deviation above it, Customer services clerks (0.84). The comparison set includes the two occupational rivals of Thewissen and Rueda (2019), skill specificity and the Walter offshorability index, extracted from their replication package and mapped to both ESS occupation codings. The gradient is 6.1 times the skill-specificity effect per standard deviation under Frey–Osborne and 1.9 times under RTI, and offshorable occupations show lower support (-0.094), the sign the benchmark study also reports. The gradient is smaller than the economic determinants estimated beside it, 67% of the education effect and 29% of the relative-income effect per standard deviation, while it exceeds the contrast between a temporary or absent contract and an unlimited one (0.026) and the household-size effect. Left–right self-placement carries the strongest coef-

ficient in the model, of which the exposure gradient is 18%. Occupational unemployment risk stays out of this model because the mechanism analysis treats it as a channel from exposure to preferences, and conditioning on the channel shrinks the gradient (Table 16). Set against these magnitudes, the post-2014 attenuation of -0.039 per standard deviation is larger in absolute size than the pooled gradient itself.

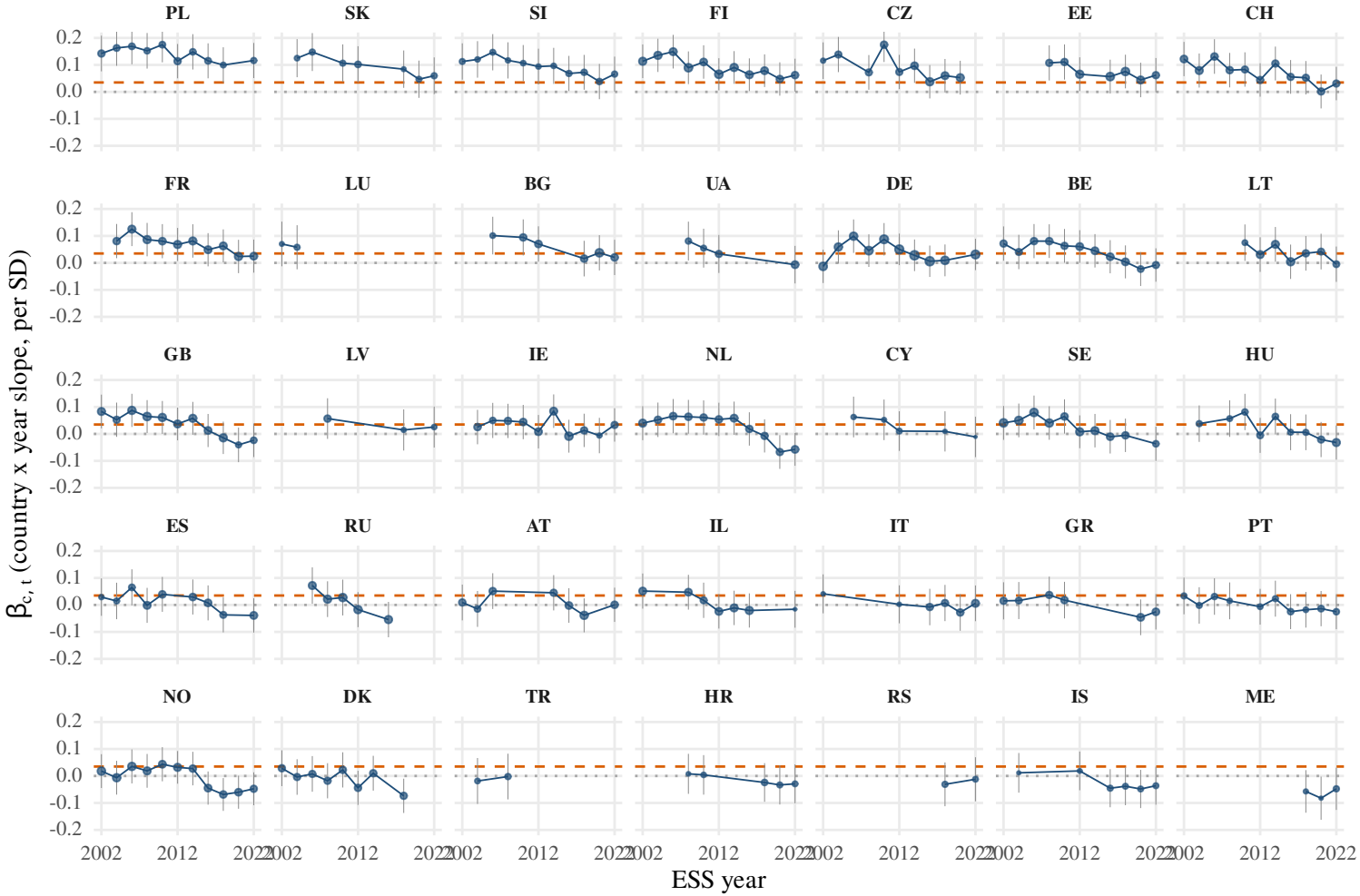
Table 3: The exposure gradient beside canonical determinants of redistribution support

	Frey–Osborne		RTI	
	Effect [95% CI]	% of mean	Effect [95% CI]	% of mean
Occupational exposure	0.034 [0.019, 0.048]	0.9	0.023 [0.012, 0.035]	0.6
Skill specificity	0.006 [0.000, 0.011]	0.1	0.012 [0.007, 0.018]	0.3
Offshorable occupation	-0.094 [-0.104, -0.084]	2.5	-0.090 [-0.101, -0.079]	2.4
Left–right self-placement	-0.193 [-0.198, -0.187]	5.0	-0.187 [-0.192, -0.181]	4.9
Relative income	-0.115 [-0.121, -0.110]	3.0	-0.122 [-0.127, -0.116]	3.2
Education	-0.051 [-0.057, -0.045]	1.3	-0.061 [-0.067, -0.055]	1.6
Temporary or no contract	0.026 [0.012, 0.039]	0.7	0.027 [0.013, 0.041]	0.7
Household size	-0.023 [-0.028, -0.018]	0.6	-0.024 [-0.030, -0.019]	0.6

One model per exposure index: redistribution support on standardized exposure, skill specificity, an offshorable-occupation indicator, left–right self-placement, contract status, household size, and the baseline control vector, with year fixed effects, a random exposure slope by country, a country-year random intercept, and ESS design weights; occupational unemployment risk is excluded. Skill specificity and the offshorability score are the occupation-level measures of the Thewissen–Rueda replication package, mapped to both ESS occupation codings; the indicator equals one where the Walter score is at least 25, their binary convention. Continuous variables are scaled to sample standard deviations; the two category contrasts are against an unlimited contract and a non-offshorable occupation. Percentages divide the absolute effect by the design-weighted outcome mean on the 1–5 scale (3.82 FO sample, 3.82 RTI sample). $N = 142,752$ (FO), $N = 132,064$ (RTI).

Figure 2 displays estimates for every country-year, ranging from negative to strongly positive values. Country-level variation exceeds temporal variation, so the association is more stable within countries than across survey waves. This pattern is consistent with the proposition that national settings condition how occupational exposure relates to redistribution preferences, though it does not by itself establish an institutional cause. The appendix provides the corresponding RTI plot.

Figure 2: Country-by-year heterogeneity in the automation–redistribution association

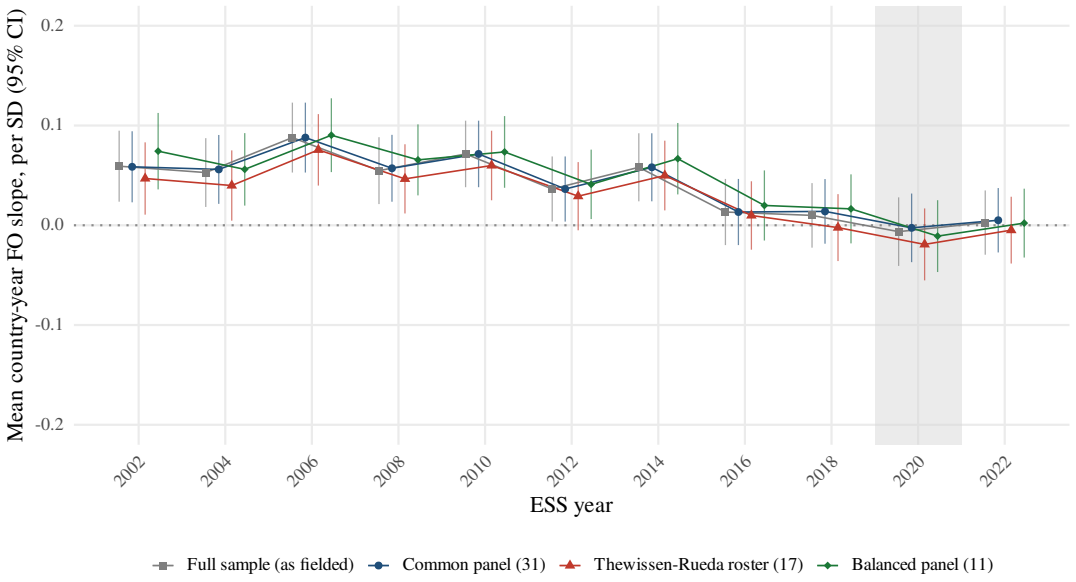


Note: Each panel is a country, ordered by mean slope; point size scales with cell N. Dashed orange = global slope; dotted line at zero. Whiskers are 95% intervals from the fitted slope surface, cropped at the panel edge where they exceed the range. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

Figure 3 separates the temporal pattern from changing country coverage. It aggregates the same fitted $\beta_{c,t}$ by year for four country rosters: all observed countries, a common panel of countries present in at least three waves with a pre-2016 anchor, the seventeen countries in the Thewissen and Rueda (2019) sample, and a balanced panel of the eleven countries present in every wave. Equal country weights make differences among curves attributable to roster composition, without re-estimating any coefficient.

The four series differ in level, with the balanced panel showing the largest slopes, but each declines after 2014, reaches a low point in the COVID-disrupted 2020 round, and partly recovers in 2022. The pattern therefore does not appear to be driven by changing country coverage. The 95% intervals combine the fixed-slope and year-component variance with averaged conditional variances for country and country-year components. The appendix reports the RTI counterpart.

Figure 3: Yearly Frey–Osborne automation–redistribution slopes under four composition assumptions

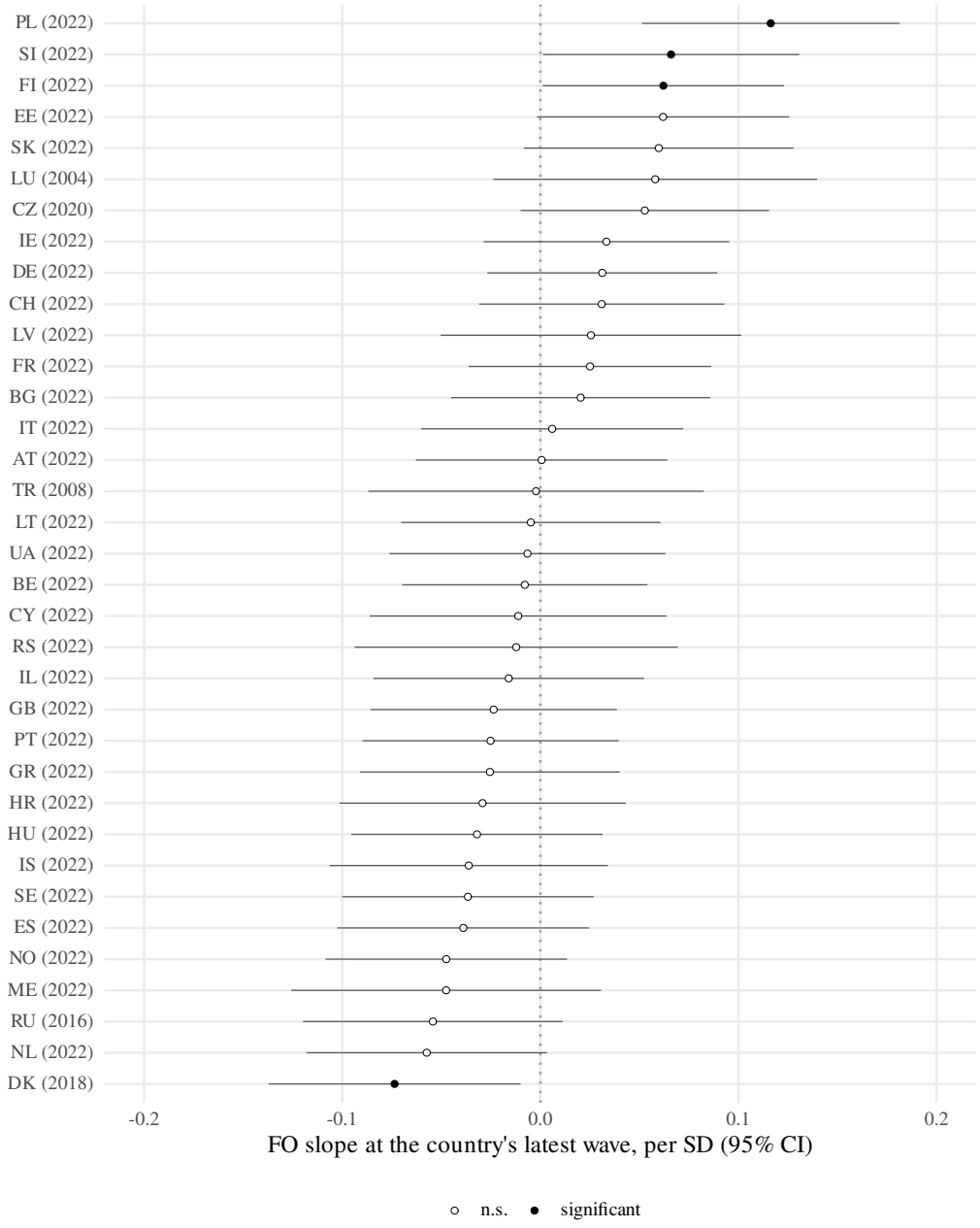


Note: Series are offset horizontally to avoid overlap; the shaded band marks the COVID-disrupted 2020 round (ESS10); dotted line at zero. Whiskers are 95% intervals from the fitted slope surface. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

Figure 4 reports the marginal effect of Frey–Osborne exposure in the latest available observation for each country. The figure labels the year because 5 of the 35 countries were last observed before 2022. Most associations are not statistically distinguishable from zero. Poland (0.116), Slovenia (0.066), and Finland (0.062) have positive significant effects, while Denmark (-0.074) has a negative significant effect. The range, from -0.07 to 0.12 per standard deviation,

suggests that the post-2014 association is specific to particular national contexts rather than a general European pattern. The corresponding RTI figure appears in the appendix.

Figure 4: Country marginal effects of Frey–Osborne exposure at the most recent available wave

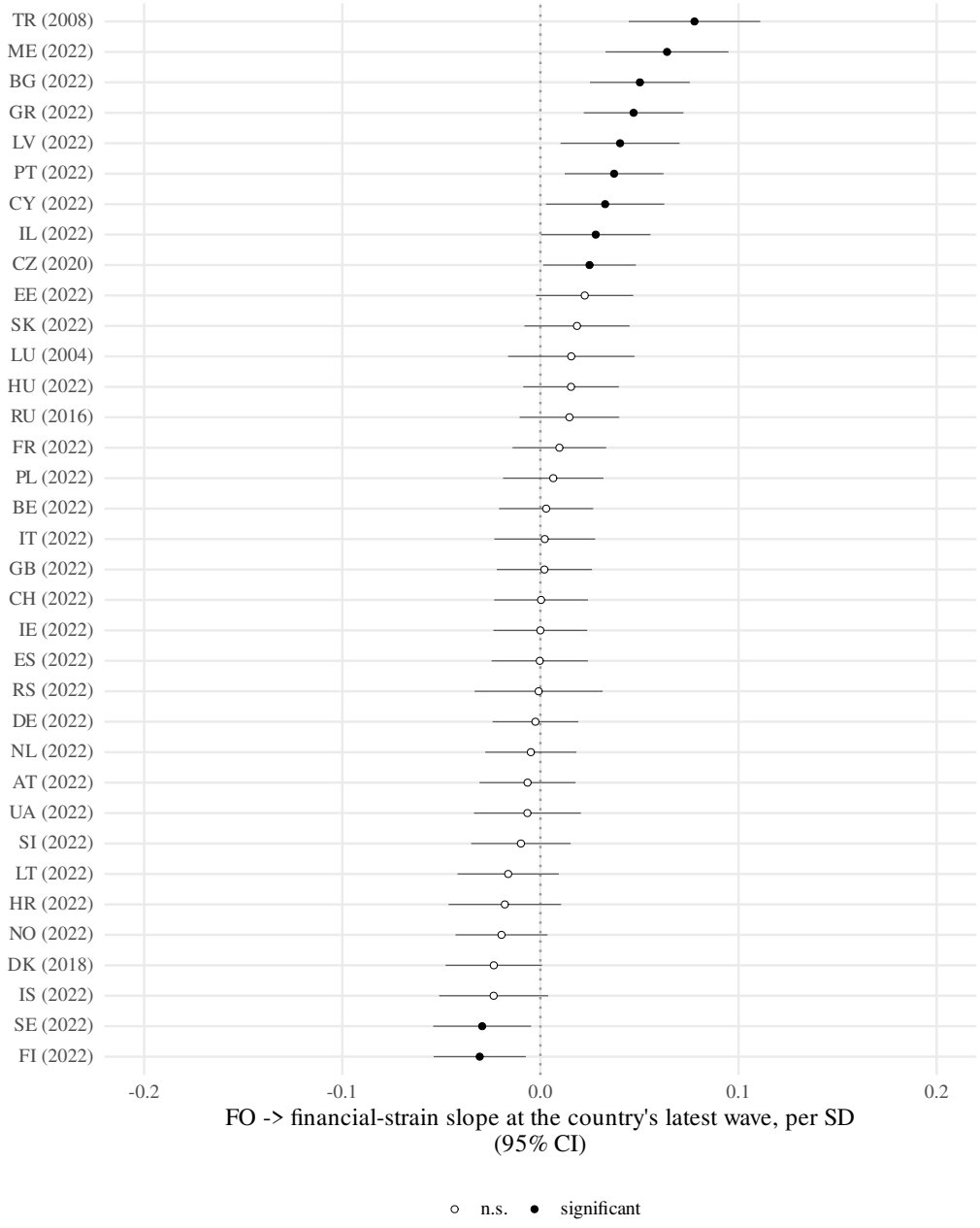


Note: Year in parentheses is each country's most recent ESS round. Whiskers are 95% intervals from the fitted slope surface; dotted line at zero. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

Figure 5 repeats the exercise with financial strain as the outcome, and the two margins diverge. At the latest wave, the exposure–strain slope is positive and significant in ten countries, concentrated in Southern and Eastern Europe, while the exposure–redistribution slope is positive in three. The countries also differ: Portugal, Greece, Bulgaria, and Latvia show strain with-

out demand, and Finland shows demand without strain, with a negative strain slope in Finland and Sweden. Under the model, the first pattern is consistent with uninsured exposure that still binds materially where demand has faded, and the second with institutions that absorb the material consequence while political demand persists. Attribution country by country is beyond the design.

Figure 5: Country marginal effects of Frey–Osborne exposure on financial strain at the most recent available wave



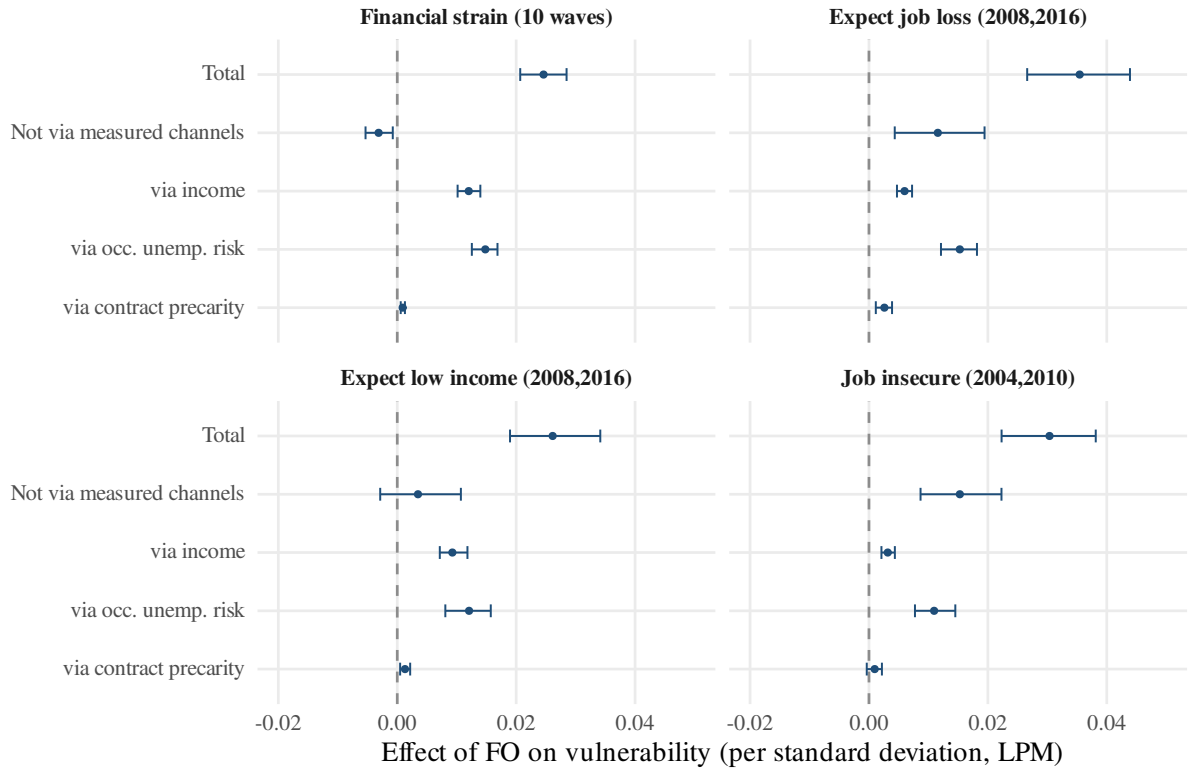
Note: Year in parentheses is each country's most recent ESS round. Whiskers are 95% intervals from the fitted strain slope surface; dotted line at zero. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

5 Mechanism Analyses: Vulnerability and Institutional Moderation

The paper's main estimand is the association between automation exposure and redistribution preferences. The mechanism analyses cover two links: from automation exposure to perceived economic vulnerability at the individual level, and from welfare-state and labor-market institutions to the strength of that exposure–vulnerability association at the country level, which would help explain why the redistribution association is stronger in some national contexts than others.

Figure 6 presents the individual-level mechanism results. Automation exposure predicts financial strain across the 10 waves in which every channel is measured. Exposure also predicts expected job loss, expected income insufficiency, and job insecurity in the rounds that include them. Appendix Table 6 reports the corresponding estimates for the 2008 and 2016 welfare-module rounds, in which the 2016 columns are the confirmatory specification and the 2008 columns are an extension under the same model. These outcomes help assess whether occupational exposure is linked to the perceived vulnerability through which it may shape redistribution preferences. The strain gradient and the redistribution gradient attenuate over the same waves, which is consistent with a common political cause and equally consistent with decay in the exposure measure itself, since a measure that tracks occupational risk less well over time would weaken both. The wave-by-wave alignment therefore corroborates the mechanism only under the assumption that the exposure indices retain their meaning across the ISCO-08 revision.

Figure 6: Automation exposure and perceived economic vulnerability across ESS waves



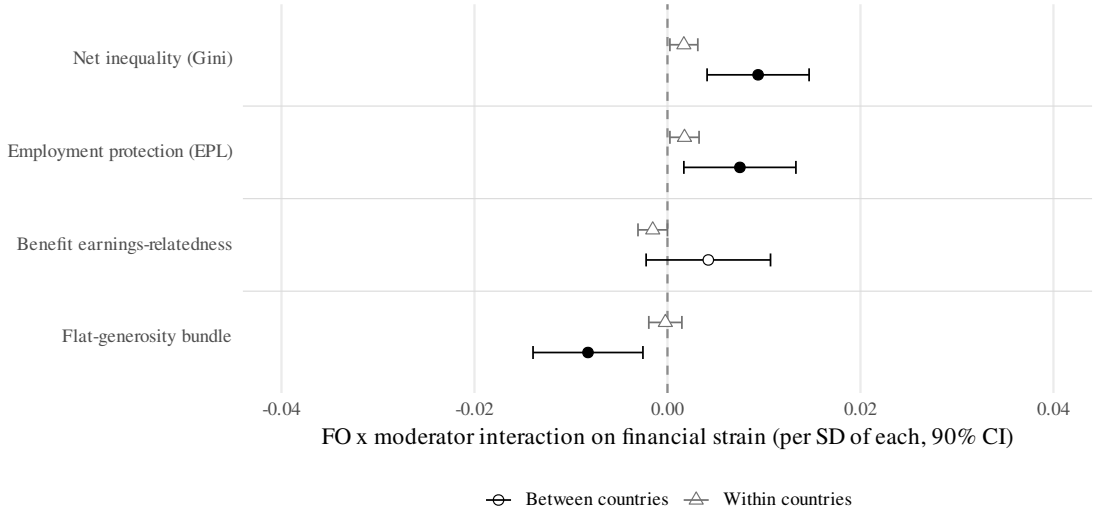
Note: Facets are the vulnerability outcomes; within each, rows decompose the association into a direct part (not via measured channels) and indirect parts through each channel (total = direct + channels). Whiskers are 95% cluster-bootstrap intervals. The decomposition is statistical, not causal mediation. Effect axis fixed at $[-0.02, +0.05]$, shared across mediation figures.

Figure 7 examines whether institutions condition the association between automation risk and financial strain, displaying the between-country and within-country components of each moderation separately. The figure reports the four theory-mapped moderators of Table 1. Because the propositions sign the redistribution gradient, the strain models are mechanism diagnostics rather than tests. The models use one-year-lagged moderators, the four institutional controls, year fixed effects, a country-year random intercept, and business-cycle adjustment, while the eleven-moderator sweep without institutional controls, the pooled single-interaction versions, and the full coefficient tables are in the appendix. Survey-year unemployment and real GDP growth enter as both main effects and interactions with exposure.

This adjustment is needed because benefits and spending expand in downturns, when the exposure–strain gradient also steepens. Without adjustment, this countercyclical co-movement obscures a buffering pattern. Net of the cycle, the moderation of strain operates between countries rather than within them. Among the focal moderators, the flat-generosity bundle buffers

the gradient (-0.008, $q = 0.064$), while employment protection (0.008, $q = 0.089$) and net inequality (0.009, $q = 0.028$) steepen it. Because the strain gradient itself is small (0.015 per standard deviation of exposure, on average across these models), these moderations are large in relative terms, between 50% and 62% of the average gradient per standard deviation of the moderator. No within-country component on strain is significant after adjustment. The two-stage diagnostic, the pooled single-interaction models, and the full sweep, which agree in direction, are in the appendix. The last result is consistent with insider–outsider dualization, in which strict protection shelters insiders while leaving exposed outsiders more precarious. Because these moderations are estimated across a median of 29 national clusters, these results should be read as evidence consistent with institutional conditioning rather than definitive tests.

Figure 7: Institutional moderation of the automation–financial-strain association

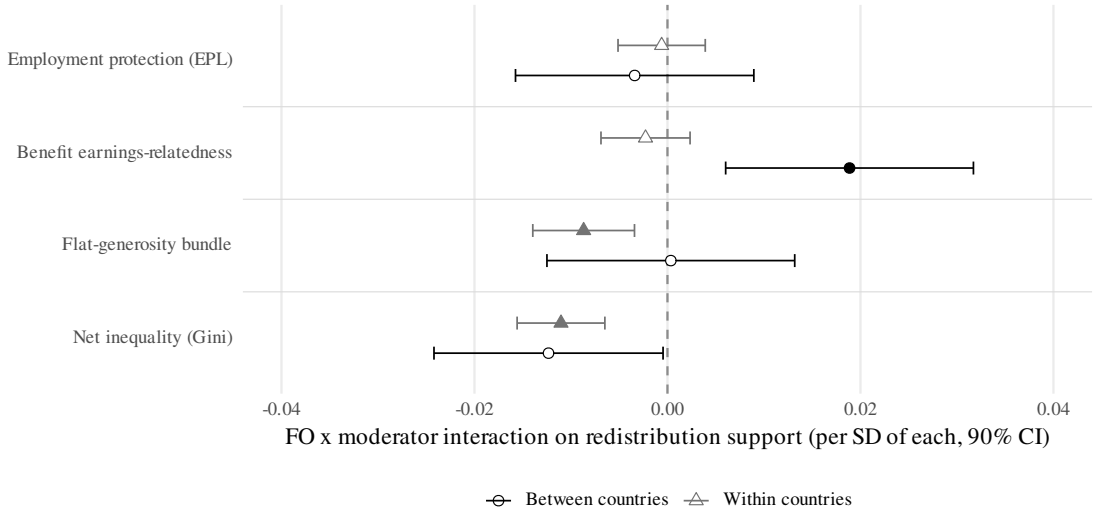


Note: Circles = between-country component (country mean of the lagged moderator); triangles = within-country (deviation from it). Filled symbols are significant at Benjamini–Hochberg $q < 0.10$, adjusted within the focal family (16 coefficients per outcome); whiskers are unadjusted 90% CIs, matched to the $q < 0.10$ criterion, so a filled symbol never crosses zero. Predicted signs: text. Effect axis fixed at $[-0.04, +0.04]$, shared across interaction figures.

The four focal tests appear against their predicted signs in Figure 8; the appendix tables report every coefficient. All three signed predictions are supported. net inequality flattens the gradient within countries (-0.011, $q = 0.001$), a 30% reduction of the average gradient (0.037) per standard deviation of the moderator. Benefit earnings-relatedness amplifies it between countries ($+0.019$, $q = 0.083$, 51% of the gradient), the sign the retention share requires, and the flat-generosity bundle buffers it within countries (-0.009, $q = 0.054$), so the two transfer routes move the gradient in the opposed directions the transfer-retention proposition predicts. Em-

ployment protection shows no effect. Its components sit near zero under Frey–Osborne, and the RTI within-country interaction is negative but not significant after adjustment (-0.006, $q = 0.115$). By Proposition 4 this is what the model predicts at the risk aversion the income results adopt, where the curvature channel is exactly zero, so the null is a correct implication rather than a failed test. The within-country components are the stronger evidence, because they are identified from changes relative to a country's own average, so time-invariant country confounds drop out. A within-country coefficient explains the temporal attenuation only if the moderator itself also moved.

Figure 8: Between- and within-country institutional moderation of the exposure–redistribution gradient



Note: Circles = between-country component (country mean of the lagged moderator); triangles = within-country (deviation from it). Filled symbols are significant at Benjamini–Hochberg $q < 0.10$, adjusted within the focal family (16 coefficients per outcome); whiskers are unadjusted 90% CIs, matched to the $q < 0.10$ criterion, so a filled symbol never crosses zero. Predicted signs: text. Effect axis fixed at $[-0.04, +0.04]$, shared across interaction figures.

Additional mechanism diagnostics, robustness checks, and full auxiliary tables appear in Appendix 8.2.

6 Discussion

The findings qualify a simple, time-invariant account of the political consequences of technological change. The association between automation risk and redistribution preferences replicates prior research in the pre-2016 period, and, in the aggregate, it all but disappears in the subsequent decade. Changing country coverage and sample composition do not account for the

decline, so something changed over time in most of the countries where the association initially held, while a few countries never showed it at all. The switch in occupation coding cannot account for this timing either. The ESS moved from ISCO-88 to ISCO-08 in 2012. Yet, under the new coding, the association remains positive and significant in both 2012 and 2014, with the 2014 slope matching the levels seen under the old regime (Appendix Figure 12). The null estimates only begin in 2016. A pure coding artifact would bind exactly at the 2012 switch and persist, rather than appearing two rounds later within an unchanged coding regime.

Individual countries illustrate the pattern (Figure 2). France shows significant positive slopes in seven of eight rounds from 2004 through 2018 and null estimates in 2020 and 2022. The same fade arrives earlier in Great Britain: significant positive slopes through 2010 become null estimates from 2012 onward, and the point estimates run negative after 2016. Finland is the counterexample, with a significant positive slope in nearly every round including 2022, while Spain never shows a significant slope in any round. Under the model, such divergence is exactly what heterogeneous insurance parameters produce. For example, the model anticipates that a shift toward flat-rate, conditional transfers—as seen in the UK's welfare reforms during the 2010s—would flatten the demand gradient, whereas encompassing, earnings-related systems like Finland's would preserve it. Although strict causal attribution of any single country's path is beyond the design, the empirical divergence aligns precisely with these theoretical mechanics.

We suspect this cross-national variation and the aggregate temporal decline are ultimately linked by a shift in the nature of technological risk itself. Routine-biased technological change in the 2000s was concentrated among clerical and manual occupations, whereas generative AI now exposes professionals with higher skills and incomes to technological precarity (Gallego and Kurer, 2022). Appendix Figure 15 documents the measured part of this shift: design-weighted routine exposure steadily declined across rounds for the benchmark roster. Because canonical indices like Frey–Osborne and RTI capture the industrial and clerical automation of the 2000s, their post-2016 attenuation serves as a methodological boundary for the field. The era of using these indices to reliably predict general left-wing political preferences has likely closed, as they no longer measure where the binding economic risk currently lies. Furthermore, as risk shifts toward higher-income professionals, the institutional mechanics of redistribution demand

change. These newly exposed workers have weaker material incentives for conventional income transfers, favoring instruments such as social investment or basic income instead (Busemeyer and Sahm, 2022).

The mechanism analyses help explain why the redistribution association concentrates where it does. Automation risk is associated with perceived vulnerability, including financial strain and expected job loss. Whether institutions buffer that vulnerability is initially harder to see. Because welfare effort expands during downturns—the very periods when the exposure–strain association steepens—unadjusted estimates can mask the buffering effect. The true institutional pattern emerges only once business-cycle dynamics enter the models.

After accounting for macroeconomic cycles, our findings align with the social-insurance account of redistribution demand (Moene and Wallerstein, 2001). High flat-transfer generosity predicts flatter exposure–strain gradients between countries, while high net income inequality predicts steeper ones. (The corresponding within-country estimates for strain are small and imprecise). Interestingly, the redistribution gradient operates on both margins: benefit earnings-relatedness amplifies it between countries, and within countries it flattens when net inequality or flat-rate generosity rises relative to a country's own average. We flag this contrast across the two outcomes rather than attempting to resolve it. Crucially, these estimates rely on observational data across a median of 29 national clusters, and while the theoretical channels are now estimated jointly, they should still be read as institutional patterns consistent with theory rather than identified causal effects.

Viewed through the model's comparative statics, these patterns become coherent. The model's core unconditional prediction—that displacement risk raises the preferred tax rate—is clearly recovered in the pre-2016 data. Because the model leaves the income-by-risk interaction unsigned at empirically plausible levels of risk aversion, we treat the benchmark positive interaction not as a universal rule, but as a joint restriction on the model's primitives. The empirical buffering effect of social spending also aligns precisely with our flat-transfer proposition. Finally, while the employment-protection results map to our curvature proposition (where lowering baseline displacement risk actually steepens the risk slope), the exact probability channel remains unmeasured here. Testing the remaining components of the bad-state income share—specifically

the re-entry wage and the expected duration of unemployment—stands as a clear priority for future measurement.

Finally, we note four scope conditions and limitations. Methodologically, our operationalization of automation risk relies on occupational-level indices (Frey–Osborne probabilities and RTI) that assume uniform task structures within broad occupations, and the harmonization between ISCO-88 and ISCO-08 introduces unavoidable crosswalk measurement error. Empirically, changing country coverage across ESS waves prevents the use of a perfectly balanced panel, though we address this via composition-adjusted common-sample models. Conceptually, our observational design restricts us from making strict causal claims; the documented associations should instead be understood as descriptive patterns fundamentally governed by their changing institutional and technological context.

7 Conclusion

Using harmonized European Social Survey data for 2002–2022, this paper reexamined the relationship between occupational automation risk and demand for redistribution. We find that the widely cited historical association between automation exposure and redistribution preferences clearly replicates in the pre-2012 era. However, this gradient all but disappears in the subsequent decade. Because this attenuation differs far more across countries than over time, the collapse of this political coalition is not driven by a uniform continental shock, but by divergent national trajectories.

The mechanism analyses reveal that political demand tracks directly with lived economic vulnerability. Automation exposure predicts actual financial strain in precisely the periods where it predicts redistribution support, and it ceases to do so when the political demand fades. Crucially, institutions determine whether this vulnerability takes root. Net of the business cycle, differences in flat-transfer generosity and income inequality govern how strictly technological exposure translates into precariousness, mapping clearly onto theories of labor market dualization between protected insiders and exposed outsiders.

Our formal insurance model grounds these empirical patterns in structural institutional dif-

ferences rather than ad-hoc psychological shifts. The model demonstrates that the core relationship—risk raising the preferred tax rate—is a robust theoretical mechanism. However, it also proves that the exact architecture of a country's welfare state strictly conditions this mechanism: whether a displaced worker's safety net scales with their prior earnings or arrives as a flat transfer mathematically dictates the shape of the risk-redistribution gradient. The observed temporal attenuation and cross-national divergence therefore reflect changing institutional parameters rather than a fundamental break in political economic logic.

These findings chart a distinct forward agenda for the discipline. The era of using routine-task indices to predict general left-wing political preferences has likely closed. As the frontier of technological risk shifts from the industrial automation of the 2000s toward generative AI, future research must deploy new exposure measures to determine whether highly skilled professionals now face the same political economic logic that previously governed routine workers. To map this new terrain, future surveys must measure the exact components of institutional buffering that our model identifies—specifically, the re-entry wage after displacement and the expected duration of unemployment—rather than relying on broad occupational aggregates. Finally, tracking these shifting political coalitions will require moving beyond cross-sectional country comparisons to leverage within-country institutional reforms over time.

References

J. Alt and T. Iversen. Inequality, labor market segmentation, and preferences for redistribution. *American Journal of Political Science*, 61(1):21–36, 2017.

Klaus Armingeon, Sarah Engler, Lucas Leemann, and David Weisstanner. Comparative political data set 1960–2023. Technical report, University of Zurich, Leuphana University Lüneburg, and University of Lucerne, 2025.

D. Autor and A. Salomons. Is automation labor-displacing? productivity growth, employment, and the labor share. Technical Report w24871, National Bureau of Economic Research, 2018.

Andrew Bell and Kelvyn Jones. Explaining fixed effects: Random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, 3(1):133–153, 2015. doi: 10.1017/psrm.2014.7.

Andrew Bell, Malcolm Fairbrother, and Kelvyn Jones. Fixed and random effects models: Making an informed choice. *Quality & Quantity*, 53:1051–1074, 2019. doi: 10.1007/s11135-018-0802-x.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.

P. Beramendi and P. Rehm. Who gives, who gains? progressivity and preferences. *Comparative Political Studies*, 49(4):529–563, 2016.

M. R. Busemeyer and A. H. Sahm. Social investment, redistribution or basic income? exploring the association between automation risk and welfare state attitudes in europe. *Journal of Social Policy*, 51(4):751–770, 2022.

M. R. Busemeyer and T. Tober. Dealing with technological change: Social policy preferences and institutional context. *Comparative Political Studies*, 56(7):968–999, 2023.

A. Elminejad, T. Havranek, and Z. Irsova. Relative risk aversion: a meta-analysis. Econstor working paper, ZBW – Leibniz Information Centre for Economics, 2022.

Malcolm Fairbrother. Two multilevel modeling techniques for analyzing comparative longitudinal survey datasets. *Political Science Research and Methods*, 2(1):119–140, 2014. doi: 10.1017/psrm.2013.24.

V. Fetscher. Explaining support for redistribution: social insurance systems and fairness. *Political Science Research and Methods*, 11(3):588–604, 2023. doi: 10.1017/psrm.2022.48.

Carl Benedikt Frey and Michael A. Osborne. The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114:254–280, 2017.

A. Gallego and T. Kurer. Automation, digitalization, and artificial intelligence in the workplace: implications for political behavior. *Annual Review of Political Science*, 25:463–484, 2022.

A. Gallego, A. Kuo, D. Manzano, and J. Fernández-Albertos. Technological risk and policy preferences. *Comparative Political Studies*, 55(1):60–92, 2022a.

A. Gallego, T. Kurer, and N. Schöll. Neither left behind nor superstar: ordinary winners of digitalization at the ballot box. *The Journal of Politics*, 84(1):418–436, 2022b.

Marco Giesselmann and Alexander W. Schmidt-Catran. Getting the within estimator of cross-level interactions in multilevel models with pooled cross-sections: Why country dummies (sometimes) do not do the job. *Sociological Methodology*, 49(1):190–219, 2019. doi: 10.1177/0081175018809150.

J. Gingrich. Did state responses to automation matter for voters? *Research and Politics*, 6(1), 2019. doi: 10.1177/2053168019832745.

Maarten Goos, Alan Manning, and Anna Salomons. Explaining job polarization: Routine-biased technological change and offshoring. *American Economic Review*, 104(8):2509–2526, 2014.

J. S. Hacker, P. Rehm, and M. Schlesinger. The insecure american: Economic experiences, financial worries, and policy attitudes. *Perspectives on Politics*, 11(1):23–49, 2013.

T. Heinrich and C. Witko. Technology threats to employment, issues, and candidate and party preferences in the united states. *Political Research Quarterly*, 75(3):797–811, 2022.

Jan Paul Heisig and Merlin Schaeffer. Why you should always include a random slope for the lower-level variable involved in a cross-level interaction. *European Sociological Review*, 35(2): 258–279, 2019. doi: 10.1093/esr/jcy053.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Michael Hout. Getting the most out of the GSS income measures. GSS Methodological Report 101, University of California, Berkeley, 2004.

Z. J. Im, N. Mayer, B. Palier, and J. Rovny. The "losers of automation": A reservoir of votes for the radical right? *Research & Politics*, 6(1):2053168018822395, 2019.

T. Iversen and D. Soskice. An asset theory of social policy preferences. *American Political Science Review*, 95(4):875–893, 2001.

K. Jeffrey. Automation and the future of work: How rhetoric shapes the response in policy preferences. QPE Working Paper 2020-10, Quantitative Political Economy Research Group, King's College London, 2020.

A. Kaihovaara and Z. J. Im. Jobs at risk? task routineness, offshorability, and attitudes toward immigration. *European Political Science Review*, 12(3):327–345, 2020.

N. C. Kakwani. Measurement of tax progressivity: an international comparison. *The Economic Journal*, 87(345):71–80, 1977.

W. Korpi and J. Palme. The paradox of redistribution and strategies of equality: Welfare state institutions, inequality, and poverty in the western countries. *American Sociological Review*, pages 661–687, 1998.

Y. Margalit. Explaining social policy preferences: Evidence from the great recession. *American Political Science Review*, 107(1):80–103, 2013.

Y. Margalit. Political responses to economic shocks. *Annual Review of Political Science*, 22: 277–295, 2019.

A. H. Meltzer and S. F. Richard. A rational theory of the size of government. *Journal of political Economy*, 89(5):914–927, 1981.

K. O. Moene and M. Wallerstein. Inequality, social insurance, and redistribution. *American Political Science Review*, 95(4):859–874, 2001.

R. L. Pahontu. Divisive jobs: three facets of risk, precarity, and redistribution. *Political Science Research and Methods*, 10(3):507–523, 2022.

R. W. Rogers. A protection motivation theory of fear appeals and attitude change1. *The journal of psychology*, 91(1):93–114, 1975.

Alexander W. Schmidt-Catran and Malcolm Fairbrother. The random effects in multilevel models: Getting them wrong and getting them right. *European Sociological Review*, 32(1):23–38, 2016.

J. D. Stephens. *The transition from capitalism to socialism*. Springer, 1979.

S. Thewissen and D. Rueda. Automation and the welfare state: Technological change as a determinant of redistribution preferences. *Comparative Political Studies*, 52(2):171–208, 2019.

A. van Hoorn. Automatability of work and preferences for redistribution. *Oxford Bulletin of Economics and Statistics*, 84(1):130–157, 2022. doi: 10.1111/obes.12460.

H. L. Wilensky. *The welfare state and equality: Structural and ideological roots of public expenditures*, volume 140. Univ of California Press, 1974.

Online Appendix

Automation Risk, Redistribution, and Economic Vulnerability

The Online Appendix, Section 8 of this paper, contains the formal insurance model of redistribution preferences and the supplementary figures and tables. It is available at

www.ignacio-urbina.com/files/automation-redistribution-appendix.pdf